



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2024-25
HALF YEARLY EXAMINATION

CLASS: X
SUBJECT: PHYSICS [SET A]

FULL MARKS: 80
TIME: 2 HOURS

*Candidates are allowed additional 15 minutes for only reading the paper.
They must NOT start writing during this time.
The intended marks for questions or parts of questions are given in brackets []
Section A is compulsory. Attempt any four questions from Section B
This paper consists of seven printed pages*

SECTION A

Question 1

[15 × 1 = 15]

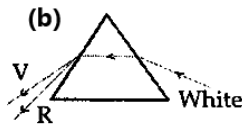
Choose the correct answers to the questions from the given options:

- (i) An object in a denser medium when viewed from a rarer medium appears to be raised. The shift is maximum for
(a) red (b) violet (c) yellow (d) green

- (ii) Which of the following diagrams is correct?



(a) (c)



(b) (d)

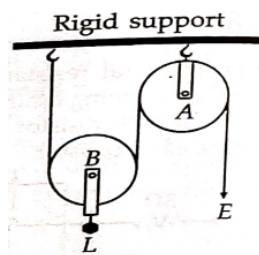


(c) (b)



(d) (a)

- (iii) The mass of an object A is double the mass of B. If both move with the same velocity, then the ratio of KE of A to B is
(a) 1:2 (b) 2:1 (c) 1:4 (d) 4:1
- (iv) An electric motor lifts an 8 kg box vertically through 2.5m in 10 s. The power produced by the motor is [g=10N/kg]
(a) 0.32W (b) 3.2W (c) 3.0W (d) 20W
- (v) In the diagram shown below the velocity ratio of the arrangement is



(a) 1

(b) 2

(c) 3

(d) none of these

(vi) Which of the following parameters is same for all electromagnetic radiations in air:
 (a) wavelength (b) frequency (c) time period (d) speed

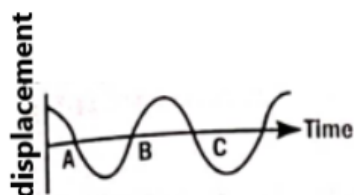
(vii) Which of the following quantities remains constant in a uniform circular motion ?
 (a) speed (b) velocity (c) acceleration (d) both velocity and speed

(viii) Assertion (A): The pitch of the sound produced by the vibrating string can be changed.

Reason (R): The frequency of the sound can be changed by changing the length, tension and thickness of the vibrating string.

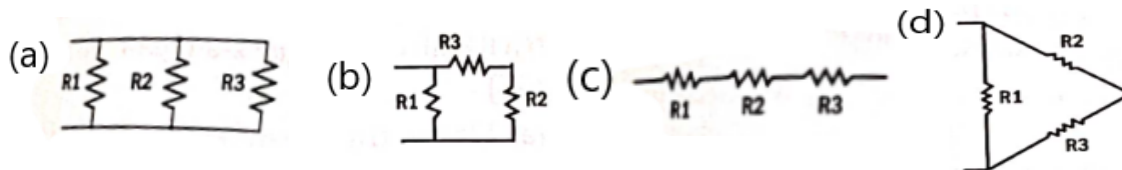
- (a) Both A and R are true and R is the correct explanation of the A.
 (b) Both A and R are true and R is not the correct explanation of the A.
 (c) A is false but R is true.
 (d) A is true but R is false.

(ix) Time period of the wave shown beside is



- (a) AB (b) BC (c) AC (d) None

(x) Three resistances R_1 , R_2 , and R_3 are connected in different fashions as shown below. Which one will exhibit the lowest equivalent resistance?



- (a) (b) (b) (a) (c) (d) (d) (c)

(xi) In the same medium where the velocity of sound remains the same, wavelength is doubled. Then its frequency will become
 (a) doubled (b) same (c) half (d) thrice

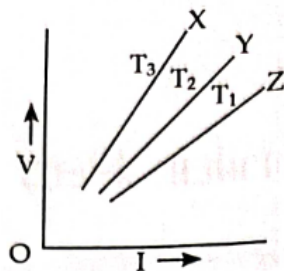
(xii) A body of mass m is moving with a uniform velocity u . A force is applied on the body due to which its velocity changes from u to v ($v > u$). How much work is being done by the force?

- (a) $\frac{1}{2} mv^2$ (b) $\frac{1}{2} m (u^2 - v^2)$ (c) $\frac{1}{2} m (v^2 - u^2)$ (d) $\frac{1}{2} m (u^4 - v^4)$

(xiii) To an astronaut in a spaceship, the earth appears to be:

- (a) white (b) red (c) blue (d) black

- (xiv) X Y and Z are three V-I graphs of a conductor at three different temperatures. The correct relation among the temperature is



- (a) $T_1 = T_2 = T_3$ (b) $T_1 > T_2 > T_3$ (c) $T_1 < T_2 < T_3$ (d) $T_1 = T_2 > T_3$
- (xv) Critical angle for diamond is 23° . The diamond shines if the angle of incidence of the light is
- (a) $> 23^\circ$ in the diamond (b) $< 23^\circ$ in the air
 (c) $< 23^\circ$ in the diamond (d) $= 23^\circ$ at the surface of separation

Question 2

- (i) Complete the following by choosing the correct answers from the bracket: [6]

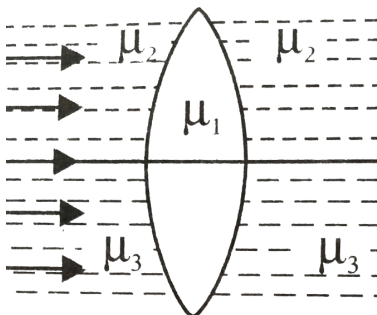
- (a) Spade lifting coal is an example of _____ [class I / class II / class III] lever.
 (b) In a block and tackle system, increase in the weight of the lower block _____ [increases/ decreases/ does not affect] the velocity ratio of the pulley system.
 (c) If the kinetic energy of a body becomes 9 times, then momentum becomes _____ [3 times/9 times/81 times]
 (d) The unit of conductance is _____ [ohm/ siemen/ ohm metre]
 (e) _____ [transverse/longitudinal] waves can travel in solids, liquids as well as gases but _____ [transverse/longitudinal] waves can travel in a solid and on the surface of a liquid.

- (ii) Draw a displacement time graph of a damped vibration. [2]

- (iii) (a) Name the radiations used in photography at night.
 (b) Give one reason for their use. [2]

Question 3

- (i) A double convex lens, made of a material of refractive index μ_1 , is placed inside two liquids of refractive indices μ_2 and μ_3 , as shown. $\mu_2 > \mu_1 > \mu_3$.



A wide, parallel beam of light is incident on the lens from the left. What type of beam will you get after refraction in medium μ_2 and medium μ_3 ? [2]

(ii) The internal resistance of a given cell is 2.5 ohm at a certain temperature. The distance between the electrodes is 5 cm. Will it decrease or increase or remain the same if: (a) temperature of the electrolyte is increased to 35°C.

(b) the distance between the electrodes increased to 10 cm. [2]

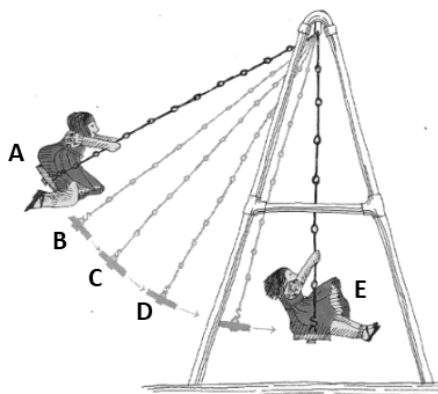
(iii) State the class of levers to which each one of the following belongs:

(a) pair of pliers

(b) sugar tongs [2]

(iv) A pitcher is placed below a running tap of water. How will the sound change if the pitcher is getting filled up? Explain. [2]

(v) What will be the change in the velocity at E if the child is raised till B or C instead of A? Explain why? [2]



(vi) Draw the circuit to verify Ohm's law. [2]

(vii) Read the following and answer the questions.

Atmospheric refraction is the phenomenon of bending of light on passing through earth's atmosphere. As we move above the surface of Earth, density of air goes on decreasing. Local conditions like temperature etc. also affect the optical density of earth's atmosphere. On account of atmospheric refraction stars appear higher than they actually are, advanced sunrise, delayed sunset, oval appearance of the sun at sunrise and sunset; stars twinkle but planets don't.

(a) What happens to the apparent length of the day due to atmospheric refraction?

(b) Which phenomenon makes the difference between the stars and the planets in terms of twinkling?

(c) What would be the colour of the sky, if there would not be any atmosphere? [3]

SECTION B
(Attempt any four questions)

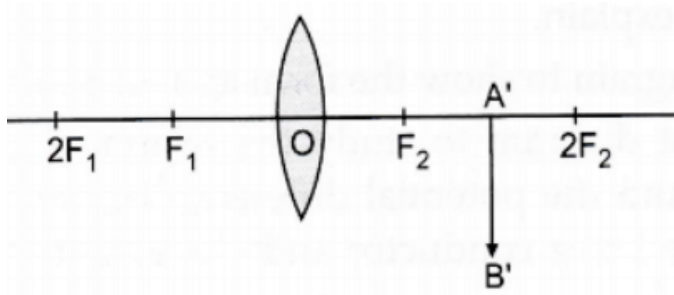
Question 4

[3+3+4]

(i) Observe the following incomplete ray diagram of an object where the image A'B' is formed after refraction through a convex lens.

(a) Complete the diagram to show the formation of the image.

(b) What would be the size of the object with respect to the size of the image?



(ii) Distinguish between ultraviolet and infrared radiation on the following points:

- (a) Heating effect
- (b) Effect on photographic plate
- (c) Fluorescence

(iii) (a) Draw the graph showing the variation of angle of deviation with the increase in angle of incidence on the prism. Mark angle of minimum deviation.

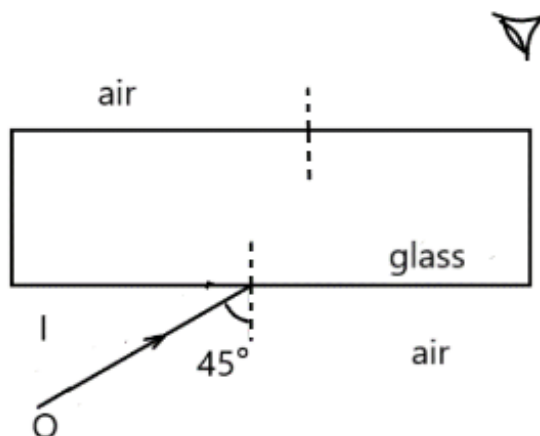
(b) The angle of an equilateral prism is 60° . If the angle of incidence in the minimum deviation position is 45° , find the angle of deviation.

Question 5

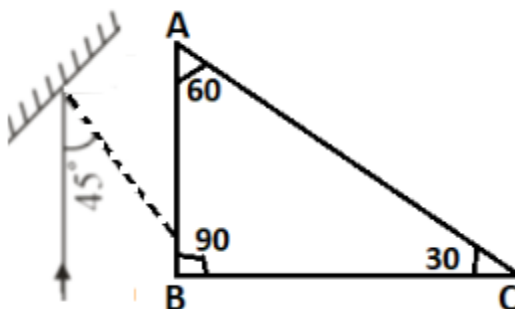
[3+3+4]

(i) A convex lens of focal length 10 cm is placed at a distance of 12 cm from a wall. How far from the lens should an object be placed so as to form its real image on the wall? What will be the magnification?

(ii) Consider a glass slab placed in air. In this case, the object (O) and the observer (eye) both in air separated by a glass slab. Complete the ray diagram to show the image (I) formation. What will be the angle of emergence?



- (iii) A ray of light is incident at an angle 45° on a plane mirror as shown below. Copy and complete the following diagrams to show the path of the ray of monochromatic light in each case as it enters and emerges out of the prism. Mark the angles wherever necessary (critical angle 42°)



Question 6

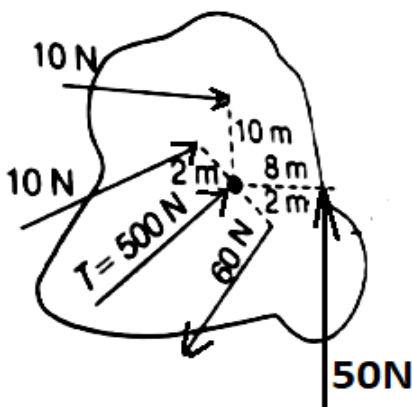
[3+3+4]

- (i) The main gate of a building is 2.6 m broad; it can be opened by applying a force of 18 N at the middle of the gate. Calculate the:
- moment of force.
 - least force that can open the gate.
- (ii) (a) Plot a graph showing the variation of resistance of a conducting wire with its length keeping the radius of the wire and its temperature as constant.
 (b) Under which condition specific resistance is equivalent to the resistance of a wire?
- (iii) (a) Distinguish between Emf of a cell and terminal voltage. [any two]
 (b) A given wire of resistance $2\ \Omega$ is stretched to double its length. What will be its new resistance?

Question 7

[3+3+4]

- (i) A rigid body is pivoted at a fixed point as shown in the following diagram. Calculate the resultant moment of force. Will the body remain in equilibrium or will rotate?



- (ii) A uniform metre rod weight 50gf balanced at 40 cm mark, a weight of 100 gf is suspended at the 5 cm mark. Where must be weight of 80 gf be suspended to balance the metre rod.

- (iii) A portion of a block and tackle system is shown in the following diagram. Complete the diagram showing the tension.



In this pulley system, if an effort of 1000 N is needed in the downward direction to raise a load of 3600 N, calculate efficiency of the system.

Question 8

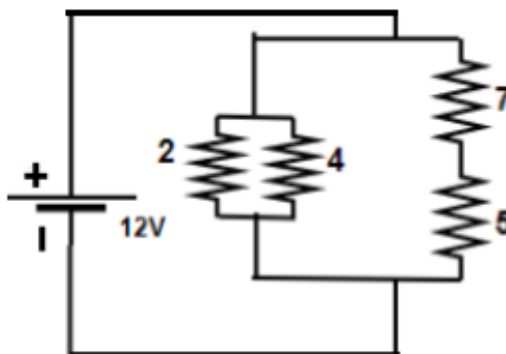
[3+3+4]

- (i) Distinguish between free and damped vibrations. [any two]. Name the subjective property of sound related to its frequency.
- (ii) (a) You can increase the loudness of a sound in various ways. Suggest any two ways.
(b) When a troop crosses a suspension bridge, the soldiers are asked to break the steps. Why?
- (iii) A man standing in a narrow valley between two cliffs gives a sharp shout. He hears two echoes, the first after 1 s and the next after 1.5 s after his shout. The speed of sound is 340 m/s. Calculate the distance between two cliffs.

Question 9

[3+3+4]

- (i) Find the equivalent resistance of the given circuit. Calculate the current in the circuit.



- (ii) Which material is used to make the heating element in an electric oven? Name two properties for their use.
- (iii) (a) An electric bulb is rated '100W, 250V'. How much current will the bulb draw if connected to a 250 V supply?
(b) Find a relation between kWh and joule.